A COPULA-BASED REGIME-SWITCHING GARCH MODEL FOR OPTIMAL FUTURES HEDGING

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The article develops a regime-switching Gumbel–Clayton (RSGC) copula GARCH model for optimal futures hedging. There are three major contributions of RSGC. First, the dependence of spot and futures return series in RSGC is modeled using switching copula instead of assuming bivariate normality. Second, RSGC adopts an independent switching Generalized Autoregressive Conditional Heteroscedasticity (GARCH) process to avoid the path-dependency problem. Third, based on the assumption of independent switching, a formula is derived for calculating the minimum variance hedge ratio. Empirical investigation in agricultural commodity markets reveals that RSGC provides good out-of-sample hedging effectiveness, illustrating importance of modeling regime shift and asymmetric dependence for futures hedging. © 2009 Wiley Periodicals, Inc. Jrl Fut Mark 29:946–972, 2009

INTRODUCTION

It is widely known that the time-varying variance-minimizing futures hedge is given by the ratio of the conditional covariance of the futures and spot returns to the conditional variance of the futures return. A wide range of Generalized Autoregressive Conditional Heteroscedasticity (GARCH) specifications have

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been applied in the literature for estimating the time-varying minimum variance hedge ratio (Baillie & Myers, 1991; Brooks, Hendry, & Persand, 2002; Byström, 2003; Gagnon & Lypny, 1995; Kroner & Sultan, 1993; Park & Switzer, 1995).

Recent studies recognize that the relationship between spot and futures returns may be characterized by regime shifts (Sarno & Valente, 2000, 2005a,b) and adopt regime-switching models to develop state-dependent dynamic hedging strategies, which are found to perform better than state-independent strategies (Alizadeh & Nomikos, 2004; Lee & Yoder, 2007a,b; Lee, Yoder, Mittelhammer, & McCluskey, 2006). Alizadeh and Nomikos (2004) and Lee et al. (2006) estimate the minimum variance hedge ratio by treating it as a time-varying coefficient. Lee and Yoder (2007a,b) estimate the minimum variance hedge ratio by estimating the conditional second moments with regime-switching GARCH models.

Although these regime-switching GARCH models have captured much of the observed behavior in the spot and futures return series, they possess some drawbacks. First, previous regime-switching GARCH hedging methods usually assume a joint normality between spot and futures return series. This may be misleading in the presence of non-normality in these returns series. Second, previous regime-switching GARCH hedging models use Gray's recombining method for solving the path-dependency problem. However, the interpretation of the variance process in Gray's approach is problematic (Hass, Mittnik, & Paolella, 2004). Third, in measuring the hedging performance, previous regime-switching hedging models derive the minimum variance hedge ratios by recombining the variances and covariance in each state. If Gray's recombining method is problematic, the minimum variance hedge ratio derived is also problematic.

This article develops a regime-switching Gumbel–Clayton (RSGC) copula GARCH model for dealing these problems. Instead of assuming bivariate normality, RSGC models the dependence structure of spot and futures return series with switching Gumbel–Clayton copula. In additions, RSGC adopts an independent switching GARCH process to avoid the path-dependency problem. Finally, a formula is derived for calculating the minimum variance hedge ratio for the hedging portfolio under independent switching assumption. RSGC is applied to corn, oats, and wheat futures contracts traded in the Chicago Board of Trade (CBOT) and cocoa futures contract traded in the New York Board of Trade (NYBOT). Results show that RSGC provides good out-of-sample hedging performance.

The remainder of the article is organized as follows. The proposed RSGC copula GARCH model is presented next. The section "Measuring Hedging Performance" derives the formula of the minimum variance hedge ratio for hedging portfolio and discusses the hedging performance measurements. This is followed by data description and empirical results. A conclusion ends the article.

REGIME-SWITCHING GUMBEL–CLAYTON (RSGC) COPULA GARCH

The previous regime-switching GARCH hedging methods (Lee & Yoder, 2007a,b) possess some drawbacks. In this section, these drawbacks will be discussed and how the proposed RSGC copula GARCH deals with these problems is also presented.[1] RSGC can be specified in the following way. Let $r_{c,t}$ and $r_{f,t}$ be the returns on the spot and futures, respectively:

$$r_{c,t} = \mu_{c,s_t} + e_{c,t,s_t} \tag{1}$$

$$r_{f,t} = \mu_{f,s_t} + e_{f,t,s_t} \tag{2}$$

where μ_{c,s_t} and μ_{f,s_t} are state-dependent mean values of spot and futures return series, and e_{c,t,s_t} and e_{f,t,s_t} are state-dependent disturbances. The state variable $s_t = \{1,2\}$ is assumed to follow a first-order, two-state Markov process and state transition probabilities are assumed to follow a logistic function such that

$$\Pr(s_t = 1 | s_{t-1} = 1) = P = \frac{\exp(p_0)}{1 + \exp(p_0)} \tag{3}$$

$$\Pr(s_t = 2 | s_{t-1} = 2) = Q = \frac{\exp(q_0)}{1 + \exp(q_0)} \tag{4}$$

where p_0 and q_0 are unconstrained parameters that can be estimated along with other unknown parameters via maximum likelihood estimation. In the previous regime-switching GARCH hedging methods (Lee and Yoder, 2007a,b), the conditional variances at time t given s_t for spot and futures returns denoted as h_{c,t,s_t} and h_{f,t,s_t} are assumed to follow a state-dependent GARCH(1,1) process

$$h_{c,t,s_t} = \gamma_{c,s_t} + \alpha_{c,s_t} e^2_{c,t-1} + \beta_{c,s_t} h_{c,t-1} \tag{5}$$

$$h_{f,t,s_t} = \gamma_{f,s_t} + \alpha_{f,s_t} e^2_{f,t-1} + \beta_{f,s_t} h_{f,t-1}. \tag{6}$$

When a recursive process is subject to regime switching, the recursive nature of the process makes the model intractable due to the dependence of the conditional variances on the entire past history of the data (Cai, 1994; Gray, 1996; Hamilton & Susmel, 1994).

To solve this path-dependency problem, previous regime-switching GARCH hedging models use Gray's (1996) recombining method to recombine the conditional variances given below:

[1]Symmetric tail dependence restricts that extremely favorable or unfavorable market conditions drive the dependency by the same magnitude, but there is mounting evidence that although financial markets are more likely to collapse or to boom together, the magnitude might not be the same or even similar (Lai, Chen, & Gerlach, 2007; Rodriguez, 2007). Therefore, Gumbel and Clayton copulas are chosen in this study because of their ability in capturing asymmetries in tail dependence. Dias and Embrechts (2004), Lai et al. (2007), and Rodriguez (2007), among others, also apply the mixture of Gumbel and Clayton copulas to the empirical finance.

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$$h_{i,t} = p_{1t}(\mu_{i,1}^2 + h_{i,t,1}) + (1 - p_{1t})(\mu_{i,2}^2 + h_{i,t,2}) - \left[p_{1t}\mu_{i,1} + (1 - p_{1t})\mu_{i,2}\right]^2 \quad (7)$$

for $i = \{c, f\}$, where p_{1t} is the probability of being in regime 1 at time t, defined as

$$\begin{aligned} p_{1t} &= \Pr(s_t = 1|\psi_{t-1}) \\ &= P\left[\frac{f_{1t-1}p_{1t-1}}{f_{1t-1}p_{1t-1} + f_{2t-1}(1 - p_{1t-1})}\right] + (1 - Q)\left[\frac{f_{2t-1}(1 - p_{1t-1})}{f_{1t-1}p_{1t-1} + f_{2t-1}(1 - p_{1t-1})}\right] \end{aligned} \quad (8)$$

and P and Q are transition probabilities defined in (3) and (4) and f_{1t-1} and f_{2t-1} are conditional densities of being in the state 1 and state 2 at time $t-1$, respectively.[2]

Although the model of Gray is attractive in that it combines Markov switching and GARCH effects and solves the estimation difficulties, its analytical intractability is a serious drawback (Hass et al., 2004). Consider a state-dependent GARCH(1,1) process,

$$h_t = \gamma + \alpha e_{t-1}^2 + \beta h_{t-1}. \quad (9)$$

If $\beta < 1$, the variance h_t can be expressed as

$$h_t = \gamma(1 - \beta)^{-1} + \alpha \sum_{i=1}^{\infty} \beta^{i-1} e_{t-i}^2 \quad (10)$$

where α reflects the magnitude of a unit shock's immediate impact on the next period's variance, β is a parameter of inertia and indicates the memory in the variance, and the total impact of a unit shock to future variance is $\alpha(1 - \beta)^{-1}$. In the regime-switching GARCH model, the relationship between the pattern with which h_t responds to shocks and the parameters α, β, and γ is far from obvious if Gray's recombining method is used because the lagged variances is replaced with the recombined variances. Moreover, in Gray's model it is possible that the variance of one regime will still be affected by shocks even if α in that regime is zero.

Hass et al. (2004) suggest an independent switching GARCH process to solve these problems. The independent switching GARCH process is specified below:

$$h_{t,1} = \gamma_1 + \alpha_1 e_{t-1,1}^2 + \beta_1 h_{t-1,1} \quad (11)$$

$$h_{t,2} = \gamma_2 + \alpha_2 e_{t-1,2}^2 + \beta_2 h_{t-1,2} \quad (12)$$

[2]For the detail derivation of this formula, please refer to Gray (1996, p. 58). For the detail descriptions of the path-dependency problem and Gray's recombining method for bivairate case, please refer to Lee and Yoder (2007b, Appendix A).

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where the subscripts 1 and 2 stand for regimes 1 and 2, respectively. This specification preserves the economic significant of the variance dynamics in each regime and admits analytical analysis of the variance process.

Following the concept of Hass et al. (2004), this study also suggests an independent switching variance dynamics for the RSGC hedging model. The volatility equations of RSGC are modeled as

$$e_{c,t,s_t} \sim N(0, h_{c,t,s_t}) \tag{13}$$

$$e_{f,t,s_t} \sim N(0, h_{f,t,s_t}) \tag{14}$$

and the conditional variance vector is specified as

$$\mathbf{h}_t = \gamma_t + \alpha_t \mathbf{e}_t^2 + \beta_t \mathbf{h}_{t-1} \tag{15}$$

where

$$\mathbf{h}_t = \begin{bmatrix} h_{c,t,1} \\ h_{c,t,2} \\ h_{f,t,1} \\ h_{f,t,2} \end{bmatrix}, \boldsymbol{\gamma}_t = \begin{bmatrix} \gamma_{c,1} \\ \gamma_{c,2} \\ \gamma_{f,1} \\ \gamma_{f,2} \end{bmatrix}, \boldsymbol{\alpha}_t = \begin{bmatrix} \alpha_{c,1} & 0 \\ \alpha_{c,2} & 0 \\ 0 & \alpha_{f,1} \\ 0 & \alpha_{f,1} \end{bmatrix}, \boldsymbol{e}_t^2 = \begin{bmatrix} e_{c,t-1}^2 \\ e_{f,t-1}^2 \end{bmatrix}, \text{ and}$$

$$\boldsymbol{\beta} = \begin{bmatrix} \beta_{c,1} & 0 & 0 & 0 \\ 0 & \beta_{c,2} & 0 & 0 \\ 0 & 0 & \beta_{f,1} & 0 \\ 0 & 0 & 0 & \beta_{f,1} \end{bmatrix} \tag{16}$$

with this specification, RSGC is free from the path-dependency problem and the using of recombining method.

Another problem of previously proposed regime-switching GARCH hedging methods (Lee & Yoder, 2007a,b) is that they impose a joint-normality restriction to the disturbance vector $\mathbf{e}_{t,s_t}$ as is given below:

$$\mathbf{e}_{t,s_t} \Big| \psi_{t-1} = \begin{bmatrix} e_{c,t,s_t} \\ e_{f,t,s_t} \end{bmatrix} \Big| \psi_{t-1} \sim BN(0, \quad \mathbf{H}_{t,st}) \tag{17}$$

where ψ_{t-1} refers to the information available at time $t-1$, BN stands for bivariate normal, and H_{t,s_t} is a time-varying, state-dependent, 2×2 positive definite conditional covariance matrix.

This restriction ignores the potential important dimension of dynamic futures hedging, the asymmetric dependence structures. The multivariate normality assumes tail independence. Financial markets, however, are more likely to collapse together during a period of stress and to boom together during a

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period of prosperity. The elliptical form of dependence inherent in a bivariate normal cannot capture the co-movement phenomenon commonly observed in the financial markets (Lai et al., 2007; Rodriguez, 2007). Instead, Gumbel and Clayton copulas exhibit, respectively, the upper tail dependence and the lower tail dependence and the mixture of Gumbel and Clayton copulas captures both these tail dependences and hence the co-movement phenomenon observed in the financial markets.

Copulas are functions that join or couple multivariate distribution functions to their one-dimension marginal distribution functions. The most important result in copula theory is Sklar's theorem, which says that it is possible to separate the univariate margins from the dependence structure represented by copula (Rodriguez, 2007). The proposed RSGC copula GARCH model captures not only the regime shifts as previous regime-switching GARCH methods do but also the non-normal and asymmetric dependence properties of spot and futures returns.

In the proposed RSGC GARCH, the marginal distributions are assumed to be state-dependent normal distributions as defined in Equations (13) and (14) and the dependence structure is modeled with the following regime-switching Gumbel–Clayton copula:

$$C(u_t, v_t|s_t, \psi_{t-1}) = \pi_{c,s_t} C_t^C(u_t, v_t;\ \delta_{t,s_t}^C|\psi_{t-1}) + (1 - \pi_{c,s_t}) C_t^G(u_t, v_t;\ \delta_{t,s_t}^G|\psi_{t-1}) \quad (18)$$

where $\pi_{c,s_t} \in [0,1]$ is a state-dependent shifting parameter of the mixture Gumbel–Clayton copula and C_t^C and C_t^G are state-dependent versions of Clayton (1978) and Gumbel (1960) copulas, respectively. Lower tail dependence increases as π_{c,s_t} goes from zero to one.

The state-dependent Clayton and Gumbel copula are defined as

$$C_t^C(u_t, v_t;\ \delta_t^C|\psi_{t-1}) = (1 + \delta_t^C)(u_t v_t)^{-1-\delta_t^C}(u_t^{-\delta_t^C} + v_t^{-\delta_t^C} - 1)^{(-1-2\delta_t^C)/\delta_t^C} \quad (19)$$

$$C_t^G(u_t, v_t;\ \delta_t^G|\psi_{t-1}) = \frac{\exp\{-\Lambda^{1/\delta_t^G}\}[\ln(u_t)\ln(v_t)]^{\delta_t^G-1}\{\Lambda^{1/\delta_t^G} + \delta_t^G - 1\}}{u_t v_t \Lambda^{2-(1/\delta_t^G)}} \quad (20)$$

where $\Lambda = [(-\ln(u_t))^{\delta_t^G} + (-\ln(v_t))^{\delta_t^G}]$, $u_t = F_c(r_{c,t}|s_t, \psi_{t-1})$, and $v_t = F_f(r_{f,t}|s_t, \psi_{t-1})$ are respectively the state-dependent conditional cumulative distribution functions of the spot and futures returns at time t, and δ_t^C and δ_t^G are parameters of Gumbel and Clayton copulas, respectively.

Patton (2006a,b) introduced the concept of conditional copula to capture the time shifts in the dependence structure. This study also allows $\delta_t^C$and δ_t^G to be time varying by defining $\delta_t^C = 2\tau_t/(1 - \tau)$ and $\delta_t^G = 1/1 - \tau_t$ (Nelsen, 1999), where τ_t is the time-varying scaled invariant dependence measure

Kendall's τ, which is given by $\tau_t = (2/\pi)\sin^{-1}(\rho_t)$ (Lai et al., 2007). The dependence process is specified as

$$\rho_t = (1 - \theta_1 - \theta_2)\rho + \theta_1\rho_{t-1} + \theta_2\phi_{t-1} \tag{21}$$

where parameters θ_1 and θ_2 are assumed non-negative and $\theta_1 + \theta_2 \leq 1$ (Tse & Tsui, 2002). The disturbance term ϕ_t is given by

$$\phi_{t-1} = \frac{\sum_{j=1}^{2}\varepsilon_{c,t-j}\varepsilon_{f,t-j}}{\sqrt{(\sum_{j=1}^{2}\varepsilon_{c,t-j}^2)(\sum_{j=1}^{2}\varepsilon_{f,t-j}^2)}} \tag{22}$$

where $\varepsilon_{c,t-1} = e_{c,t-1}/\sqrt{h_{c,t}}$ and $\varepsilon_{f,t-1} = e_{f,t-1}/\sqrt{h_{f,t}}$ are standardized residuals of spot and futures returns.

Equations (1), (2), (13)–(16), and (18)–(22) constitute the specification of the RSGC model. The unknown parameters in RSGC are $\Theta = \{p_0, q_0, \mu_{c,s_t}, \mu_{f,s_t}, \gamma_{c,s_t}, \gamma_{f,s_t}, \alpha_{c,s_t}, \alpha_{f,s_t}, \beta_{c,s_t}, \beta_{f,s_t}, \pi_{,s_t}, \theta_1, \theta_2, \rho\}$ for $s_t = 1, 2$, which can be estimated by maximizing the following log-likelihood function[3]:

$$L(\Theta) = \sum_{t=1}^{T}\log[q(r_{c,t}, r_{f,t}|\psi_{t-1})] \tag{23}$$

where the likelihood of each observation can be written as

$$\begin{aligned} q(r_{c,t}, r_{f,t}|\psi_{t-1}) &= \sum_{i=1}^{2} q(r_{c,t}, r_{f,t}, s_t = i|\psi_{t-1}) \\ &= \sum_{i=1}^{2} C(u_t, v_t|s_t = i, \psi_{t-1}) \times f(r_{c,t}|s_t = i, \psi_{t-1}) \\ &\quad \times g(r_{f,t}|s_t = i,\psi_{t-1}) \times p(s_t = i|\psi_{t-1}) \end{aligned} \tag{24}$$

with C the copula density and f and g the marginal densities of spot and futures returns, respectively.[4]

[3]When the copula and marginal parameters change simultaneously according to a Markov Switching process, the two-step approach cannot be used. All parameters must be estimated simultaneously (Rodriguez, 2007).

[4]The likelihood function consists of a sum, which creates problems for working with log-likelihood functions. Hamilton (1994, p. 686) suggests that the maximum likelihood estimates of parameters can be achieved using the numerical methods. Hamilton (1994, p. 689) mentions that the mixture density has the property that a global maximum of the log likelihood does not exist and a singularity arises whenever one of the distributions is imputed to have a mean exactly equal to one of the observations with no variance. However, such singularities do not pose a major problem in practice, because numerical maximization procedures typically converge to a reasonable local maximum rather than a singularity. The largest local maximum is described as the maximum likelihood estimate. If a numerical maximization algorithm becomes stuck at a singularity, one satisfactory solution is simply to ignore the singularity and try again with different starting values. To save space, the detail procedure and flowchart for evaluating the likelihood function are not shown in this article but are available on request.

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MEASURING HEDGING PERFORMANCE

Hedging performance is evaluated from both a risk-minimization and a utility standpoint in this article. Let χ_t be the estimated optimal hedge ratios derived from various hedging strategies. The estimated time-varying minimum variance hedge ratio denoted as χ_t is given by

$$\chi_t = \frac{Cov(r_{c,t}, r_{f,t}|\psi_{t-1})}{Var(r_{f,t}|\psi_{t-1})}. \tag{25}$$

If the hedge ratio is subjected to regime shifts, the recombined covariance and variance are found first and the optimal hedge ratio is then calculated with Equation (25). As explained in the section "Regime-Switching Gumbel–Clayton (RSGC) Copula GARCH," this recombining method possesses some drawbacks. In the RSGC model, based on the assumption of independent switching, the time-varying minimum variance hedge ratio can be derived without recombining procedure and is given by

$$\chi_t = \frac{p_{1t}^2 Cov(r_{c,1,t}, r_{f,1,t}) + (1 - p_{1t})^2 Cov(r_{c,2,t}, r_{f,2,t})}{p_{1t}^2 Var(r_{f,1,t}) + (1 - p_{1t})^2 Var(r_{f,2,t})}.^{5} \tag{26}$$

From a risk-minimization standpoint, a hedger chooses a hedging strategy to minimize the variance of the hedged portfolio return or equivalently to maximize the variance reduction of a hedging strategy compared to the unhedged position.

Because dynamic hedging strategies are potentially more costly implement than static models as frequent rebalancing of the hedged portfolio is required, hedging effectiveness is more appropriately assessed by considering the economic benefits measured with utility functions. Consider a hedger with a mean-variance expected utility function (Alizadeh & Nomikos, 2004; Gagnon, Lypny, & McCurdy, 1998; Kroner & Sultan, 1993; Lafuente & Novales, 2003; Lee et al., 2006).

[5]This can be easily proved as follows. Let r_p be the hedging portfolio return which is given by

$$\begin{aligned} r_p &= p_1(\text{portfolio return in state 1}) + (1 - p_1)(\text{portfolio return in state 2}) \\ &= p_1(r_{c,1} - \chi r_{f,1}) + (1 - p_1)(r_{c,2} - \chi r_{f,2}) \end{aligned}$$

where $r_{c,i}$ and $r_{f,i}$ are the returns spot and futures would have in the state i, respectively. We would like to choose χ such that the variance of the portfolio return is minimized.

$$\min_{\chi} Var(r_p) = \min_{\chi} \left\{ \begin{array}{l} p_1^2[Var(r_{c,1}) - 2\chi Cov(r_{c,1}, r_{f,1}) + \chi^2 Var(r_{f,1})] \\ +(1 - p_1)^2[Var(r_{c2}) - 2\chi Cov(r_{c,2}, r_{f,2}) + \chi^2 Var(r_{f,2})] \\ +2p_1(1 - p_1)\, Cov(r_{c,1} - \chi r_{f,1}, r_{c,2} - \chi r_{f,2}) \end{array} \right\}$$

Deriving the above equation with respect to χ and using the assumption of independent switching gives Equation (26).

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$$E[U(r_{p,t})|\psi_{t-1}] = E[r_{p,t}|\psi_{t-1}] - \xi Var(r_{p,t}|\psi_{t-1}) \tag{27}$$

where ξ is the coefficient of absolute risk aversion, E stands for expectation operator, and $r_{p,t}$ is the return from the hedged portfolio calculated as $r_{p,t+1} = r_{c,t+1} - \chi_t r_{f,t+1}$. A dynamic hedging strategy is considered to be superior to a static ordinary least square (OLS) method if it has higher expected utility net of transaction costs.

In addition to measuring the economic significance of dynamic hedging strategies with utility function, the statistical significance of the variance reduction is also tested with the Diebold–Mariano (DM) statistic (Corradi, Swanson, & Olivetti, 2001; Diebold & Mariano, 1995). DM statistic evaluates explicit tests of the null hypothesis of difference in the accuracy of two competing forecasts. Define the variances of the hedging portfolio returns of the benchmark and alternative model as $f_{b,t} = r_{c,t} - \hat{\beta}_{b,t} r_{f,t}$ and $f_{a,t} = r_{c,t} - \hat{\beta}_{a,t} r_{f,t}$, respectively, where $\hat{\beta}_{b,t}$ is the hedge ratio estimated using benchmark and $\hat{\beta}_{a,t}$ is the hedge ratio estimated using alternative model. The DM statistic is given by

$$DM = N^{-1/2} \frac{\sum_{t=R+1}^{T} (f_{b,t+1} - f_{a,t+1})}{\hat{\sigma}_N} \tag{28}$$

where R denotes the length of estimation period, N is the length of the prediction period, T is the sample size and $\hat{\sigma}_N^2$ is defined as

$$\hat{\sigma}_N^2 = \frac{1}{N} \sum_{t=R}^{T-1} (f_{b,t+1} - f_{a,t+1})^2 + \frac{2}{N} \sum_{j=1}^{l_N} w_j \sum_{t=R+j}^{T-1} (f_{b,t+1} - f_{a,t+1})(f_{b,t+1-j} - f_{a,t+1-j}) \tag{29}$$

where $w_j = 1 - (j/(l_N + 1) l_N = o(N^{1/4}))$.

The null of equal variance is

$$H_0 = E[f_{b,t+1} - f_{a,t+1}] = 0 \tag{30}$$

whereas the alternative is

$$H_A = E[f_{b,t+1} - f_{a,t+1}] \neq 0. \tag{31}$$

Diebold and Mariano (1995) show that under the null of equal predictive accuracy, the DM statistic converges in distribution to standard normal denoted as

$$DM \xrightarrow{d} N(0, 1).$$

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Hence, we reject the null of equal variance at the 5% level if $|DM| > 1.96$. One-sided tests can also be computed.[6] The null that benchmark is not outperformed by alternative model is not rejected at the 5% level if $DM < -1.645$.

DATA DESCRIPTION AND EMPIRICAL RESULTS

Corn, oats, and wheat nearby futures contract traded in the CBOT and cocoa nearby futures contract traded in the NYBOT are investigated in this study. Both spot and futures rates were collected from Datastream from January 1991 to December 2007 and the spot and futures data are Wednesday's closing price. The returns of each price series are computed as the changes in the natural logarithms of prices multiplied by 100. The sample is split into two periods: pre-2000 (from January 1991 to December 1999) and post-2000 (from January 2000 to December 2007). Both sub-periods and full-sample period are investigated in this study.

Summary statistics of corn, oats, wheat, and cocoa data are shown in Table I. For the pre-2000 period, most of the mean returns are negative and for the post-2000 period, all of the mean returns are positive with magnitude higher than that in pre-2000 period. This leads to a positive mean return for the whole-sample period. However, all spot and futures returns are small. The largest return is the return of wheat futures in post-2000 period, which is only 0.32%. The unconditional volatilities indicate that the post-2000 period is more volatile than pre-2000 period for all agricultural commodities investigated in this study. This is consistent with the fact that higher return accompanies higher risk. Skewness, leptokurtosis, and significant Jarque–Bera statistics are present in all data, implying that the unconditional distributions of spot and futures returns are asymmetric, fat-tailed, and non-Gaussian.

Parameter estimates from alternative models are presented in Table II. In-sample estimation of all models was conducted using data from January 1991 to December 2006; the data of the most recent year are used for out-of-sample analysis. The parameters are estimated by maximizing the log-likelihood functions using numerical constrained optimization procedure (Newton–Raphson method) in GAUSS. Shown in the last row of Table II, LRT reports the statistics of standard likelihood ratio test for nested model and Vuong's test statistics (Vuong, 1989) for non-nested model. In this study, RSGC nested within it the GC model but not the CC or VC GARCH models. Vuong's test for selection of non-nested models is related to the classical likelihood ratio-based test and uses Kullback–Leibler Information criterion to measure the closeness

[6]Alternative test of the null that benchmark is not outperform by alternative model is also available. In this case, the null should be stated with the $\leq$ sign and the alternative with the $>$ sign. The statistics diverges to minus infinity whenever the null holds (Corradi et al., 2001).

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TABLE I
Summary Statistics for Spot and Futures Returns of Full-Sample and Two Sub-Sample Periods

	1991–1907		*1991–1999*		*2000–2007*	
	Spot	*Futures*	*Spot*	*Futures*	*Spot*	*Futures*
Corn						
Mean	0.063	0.075	−0.053	−0.027	0.195	0.191
Maximum	15.978	13.799	15.978	13.799	12.647	12.447
Minimum	−14.833	−15.310	−14.833	−15.310	−14.653	−13.707
Standard deviation	3.509	3.372	3.183	3.027	3.842	3.722
Skewness	−0.126	0.186	−0.201	0.115	−0.108	0.193
Kurtosis	5.027	4.531	6.233	5.660	4.132	3.688
Jarque–Bera	153.96**	91.57**	207.41**	139.31**	23.08**	10.81**
Oats						
Mean	0.107	0.117	−0.003	0.000	0.231	0.249
Maximum	20.580	16.115	13.873	16.115	20.580	15.415
Minimum	−22.868	−14.888	−13.968	−12.783	−22.868	−14.888
Standard deviation	4.247	4.427	3.890	4.119	4.618	4.752
Skewness	−0.054	0.162	−0.125	0.241	−0.028	0.082
Kurtosis	6.598	4.196	5.247	4.310	7.072	3.998
Jarque–Bera	478.27**	56.71**	99.91**	38.05**	288.13**	17.79**
Wheat						
Mean	0.131	0.144	−0.035	−0.014	0.318	0.321
Maximum	20.576	16.954	20.576	16.954	13.861	13.272
Minimum	−24.997	−18.332	−24.997	−18.332	−12.671	−10.559
Standard deviation	4.196	3.682	3.925	3.378	4.479	3.993
Skewness	0.018	0.202	−0.100	0.017	0.079	0.288
Kurtosis	5.098	4.249	7.740	6.109	3.188	2.953
Jarque–Bera	162.58**	63.63**	439.78**	188.95**	1.05	5.79
Cocoa						
Mean	0.099	0.065	0.003	−0.065	0.209	0.211
Maximum	33.841	20.191	33.841	19.133	19.129	20.191
Minimum	−20.371	−14.129	−20.371	−10.182	−18.539	−14.129
Standard deviation	3.983	4.095	3.763	3.701	4.219	4.497
Skewness	0.963	0.286	2.238	0.584	−0.077	0.062
Kurtosis	13.232	4.737	23.248	5.342	6.003	4.205
Jarque–Bera	4,002.02**	123.51**	8,402.77**	133.83**	157.06**	25.47**

Note. **Indicates significance at the 1% level and returns are calculated as the differences in the logarithm of price multiplied by 100.

of the model to the true data generating process. The test is used to decide which model is performing better than the other one in explaining the data under consideration. Vuong's non-nested likelihood ratio test statistics is given by

$$LR = \frac{\sum_{t=1}^{n} [\ln f_A(y_t) - \ln f_B(y_t)]}{\sqrt{n}\hat{S}_n} \xrightarrow{d} N(0, 1) \tag{32}$$

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TABLE II
Estimates of Unknown Parameters of Alternative Models' Data Period Are from January 1991–December 2006

	Corn				*Oats*			
	CC	*VC*	*GC*	*RSGC*	*CC*	*VC*	*GC*	*RSGC*
p_0				0.751				1.219
				(0.285)**				(0.428)**
q_0				−1.200				0.046
				(0.440)**				(0.300)
μ_{c1}	0.279	−0.020	0.037	−0.215	0.124	0.449	0.451	0.004
	(0.100)**	(0.039)	(0.035)	(0.125)*	(0.137)	(0.004)**	(0.010)**	(0.025)
μ_{c2}				0.845				0.056
				(0.334)**				(0.186)
μ_{f1}	0.286	−0.104	−0.104	−0.538	0.151	0.446	0.448	−0.053
	(0.102)**	(0.000)**	(0.000)**	(0.123)**	(0.110)	(0.066)**	(0.019)**	(0.060)
μ_{f2}				1.542				0.323
				(0.453)**				(0.211)
γ_{c1}	0.837	1.086	1.093	0.106	16.340	16.343	10.607	0.318
	(0.194)**	(0.252)**	(0.310)**	(0.143)	(0.934)**	(1.296)**	(2.782)**	(0.048)**
γ_{c2}				2.553				40.251
				(1.461)*				(5.962)**
γ_{f1}	0.786	0.950	1.229	0.106	6.977	6.818	11.465	0.318
	(0.233)**	(0.271)**	(0.378)**	(0.123)	(1.777)**	(1.951)**	(4.872)**	(0.060)**
γ_{f2}				3.732				16.409
				(2.060)*				(25.817)
α_{c1}	0.130	0.134	0.077	0.044	0.089	0.101	0.061	0.002
	(0.022)**	(0.024)**	(0.018)**	(0.012)**	(0.027)**	(0.033)**	(0.024)**	(0.001)
α_{c2}				0.222				0.134
				(0.063)**				(0.132)
α_{f1}	0.075	0.075	0.053	0.029	0.117	0.105	0.094	0.005
	(0.015)**	(0.016)**	(0.017)**	(0.009)**	(0.036)**	(0.034)**	(0.050)*	(0.003)
α_{f2}				0.160				0.093
				(0.072)*				(0.171)
β_{c1}	0.801	0.774	0.834	0.865	0.000	0.026	0.339	0.986
	(0.030)**	(0.038)**	(0.036)**	(0.025)**	(0.021)	(0.049)	(0.164)**	(0.004)**
β_{c2}				0.778				0.000
				(0.065)**				(0.084)
β_{f1}	0.852	0.836	0.833	0.914	0.533	0.578	0.368	0.940
	(0.031)**	(0.035)**	(0.045)**	(0.020)**	(0.106)**	(0.108)**	(0.259)	(0.051)**
β_{f2}				0.731				0.636
				(0.096)**				(0.515)
θ_1		0.000	0.187	0.345		0.731	0.728	0.674
		(0.004)	(0.366)	(0.287)		(0.030)**	(0.041)**	(0.097)**
θ_2		0.200	0.076	0.085		0.195	0.167	0.125
		(0.048)**	(0.029)**	(0.035)**		(0.026)**	(0.036)**	(0.048)**
ρ		0.859	0.931	0.943		0.695	0.950	1.000
		(0.011)**	(0.006)**	(0.012)**		(0.058)**	(0.015)**	(0.058)**
π_1			0.445	0.235			0.507	0.597
			(0.043)**	(0.077)**			(0.046)**	(0.075)**
π_2				0.385				0.354
				(0.106)**				(0.090)**
LRT	51.57	45.98	164.03		33.94	34.78	266.88	

(*Continued*)

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TABLE II (*Continued*)

	Wheat				*Cocoa*			
	CC	*VC*	*GC*	*RSGC*	*CC*	*VC*	*GC*	*RSGC*
p_0				1.003				0.545
				(0.548)**				(0.572)
q_0				2.117				2.977
				(0.359)**				(0.456)**
μ_{c1}	0.109	0.383	0.391	(0.048	0.098	0.175	0.246	−0.023
	(0.108)	(0.086)**	(0.121)**	(0.217)	(0.101)	(0.092)*	(0.046)**	(0.076)
μ_{c2}				0.479				−0.149
				(0.274)*				(0.232)
μ_{f1}	0.100	0.378	0.301	(0.145	0.068	0.168	0.142	−0.082
	(0.125)	(0.094)**	(0.121)**	(0.207)	(0.081)	(0.096)*	(0.095)	(0.086)
μ_{f2}				0.596				0.647
				(0.264)*				(0.031)**
γ_{c1}	5.693	7.746	1.714	0.519	0.588	0.227	0.587	0.042
	(1.620)**	(1.726)**	(0.749)*	(0.666)	(0.219)**	(0.109)**	(0.005)**	(0.055)
γ_{c2}				0.512				0.303
				(0.904)				(2.876)
γ_{f1}	7.750	11.778	0.007	1.369	0.095	0.096	0.027	0.021
	(3.242)**	(0.850)**	(0.020)	(0.721)	(0.060)	(0.056)*	(0.041)	(0.039)*
γ_{f2}				0.512				0.303
				(0.264)*				(0.031)**
α_{c1}	0.195	0.211	0.028	0.005	0.040	0.027	0.040	0.020
	(0.041)**	(0.042)**	(0.013)*	(0.007)	(0.011)**	(0.008)**	(0.008)**	(0.005)**
α_{c2}				0.000				0.011
				(0.020)				(0.067)
α_{f1}	0.114	0.127	0.002	0.000	0.021	0.023	0.014	0.020
	(0.037)**	(0.041)**	(0.002)	(0.007)	(0.006)**	(0.007)**	(0.004)**	(0.006)*
α_{f2}				0.032				0.004
				(0.057)				(0.065)
β_{c1}	0.486	0.349	0.882	0.937	0.916	0.956	0.919	0.963
	(0.112)**	(0.113)**	(0.047)**	(0.077)**	(0.025)**	(0.015)**	(0.007)**	(0.010)**
β_{c2}				0.971				0.989
				(0.030)**				(0.034)**
β_{f1}	0.292	0.000	0.998	0.832	0.973	0.970	0.984	0.971
	(0.261)**	(0.044)	(0.002)**	(0.062)**	(0.008)**	(0.008)**	(0.005)**	(0.009)**
β_{f2}				0.963				0.991
				(0.094)**				(0.019)**
θ_1		0.912	0.890	0.912		0.938	0.939	0.937
		(0.019)**	(0.039)**	(0.044)**		(0.016)**	(0.015)**	(0.015)**
θ_2		0.052	0.031	0.030		0.047	0.049	0.051
		(0.011)**	(0.010)**	(0.016)*		(0.013)**	(0.013)**	(0.012)**
ρ		0.831	0.900	0.885		0.789	1.000	1.000
		(0.027)**	(0.014)**	(0.019)**		(0.080)**	(0.120)**	(0.217)**
π_1			0.511	0.386			0.316	0.263
			(0.050)**	(0.083)**			(0.066)**	(0.109)**
π_2				0.600				0.218
				(0.104)**				(0.455)
LRT	62.02	60.73	117.68		92.61	63.91	251.05	

Note. (1) Figures in parentheses are standard errors and *and **indicate significance at the 5% level and 1% level, respectively. (2) State 2 is the high-volatility state and state 1 is the low-volatility state. (3) LRT stands for the likelihood ratio test. Because GC is nested to RSGC, the LRT statistics is given by $LR = -2(\ln(L_r) - \ln(L_u))$, where L_y and L_u are likelihood values of restricted (GC) and unrestricted (RSGC) models, respectively. The critical value at 1% with a degree-of-freedom equal to 11 for the LRT is 24.72. As CC and VC are not nested to RSGC, LRT reports the statistics of the non-nested likelihood ratio test (Vuong, 1989). Vuong's non-nested likelihood ratio test statistics is given by

$$LR = \frac{\sum_{t=1}^{n}[\ln f_A(y_t) - \ln f_B(y_t)]}{\sqrt{n}\hat{S}_n} \xrightarrow{d} N(0,1)$$

where $f_A(y_t)$ and $f_B(y_t)$ are the conditional densities at observation t of competing models A and B, respectively, n is the number of observations, and $\hat{S}_n$ is the variance of the test statistics given by $\hat{S}_n^2 = \frac{1}{n}\sum_{t=1}^{n}[\ln f_A(y_t) - \ln f_B(y_t)]^2 - \left\{\frac{1}{n}\sum_{t=1}^{n}[\ln f_A(y_t) - \ln f_B(y_t)]\right\}^2$ The test statistics converges in distribution to standard normal under the null hypothesis that the competing models are equally close to the true data generating process. The critical value at 1% for the LRT is 2.33.

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where $f_A(y_t)$ and $f_B(y_t)$ are the conditional densities at observation t of competing model A and model B, respectively, n is the number of observations, and $\hat{S}_n$ is the variance of the test statistics given by

$$\hat{S}_n^2 = \frac{1}{n}\sum_{t=1}^{n}[\ln f_A(y_t) - \ln f_B(y_t)]^2 - \left\{\frac{1}{n}\sum_{t=1}^{n}[\ln f_A(y_t) - \ln f_B(y_t)\right\}^2. \quad (33)$$

The test statistics converges in distribution to standard normal under the null hypothesis that the competing models are equally close to the true data generating process.

For the standard likelihood ratio test, the critical value at 1% level with degree-of-freedom equal to 11 is 24.72. The standard likelihood ratio test statistics are 164.03, 266.88, 117.68, and 251.05 for corn, oats, wheat, and cocoa data, respectively. All of these values are larger than 24.72, revealing that RSGC performs better than GC in explaining the data. Similar results are also hold for the Vuong's test. For all data investigated, non-nested Vuong's test statistics are larger than 2.33, the critical value at 1% level, implying that RSGC performs better than CC and VC in explaining the data.

All conditional mean μs estimated by using state-independent models are small and less than 0.4%. This is consistent with the small average return reported in the summary statistics table. However, when we take account of regime shifts, the conditional mean return could be as large as 1.54% such as that in the corn data. This is because the conditional mean return in a state-dependent model is the weighted average of the conditional mean returns in each regime, weighted by the regime probabilities. The figure 1.54% simply implies the upper bound of the conditional mean return not the estimated conditional mean return. Generally speaking, conditional mean return is larger in the high-volatility state (regime 2) than in the lower volatility state (regime 1). The only exception is the spot data of cocoa.

In the volatility equation, β indicates the memory in the variance and $\alpha(1-\beta)^{-1}$ represents the total impact of a unit shock to future variance. For corn data, the βs are 0.865 and 0.914 for spot and futures return series in low-volatility state (regime 1) and 0.778 and 0.731 for spot and futures return series in high-volatility state (regime 2). This shows that volatilities in the low-volatility state have higher shock persistence than that in the high-volatility state. Oats data have similar results. However, for wheat and cocoa data, the volatilities in the high-volatility state tend to have higher shock persistence than in the low-volatility state. It is also found that for the spot data, corn, and cocoa in the high-volatility state have the highest total impact of a unit shock to future variance. As for the futures data, wheat in the high-volatility state has the highest total impact of a unit shock to future variance.

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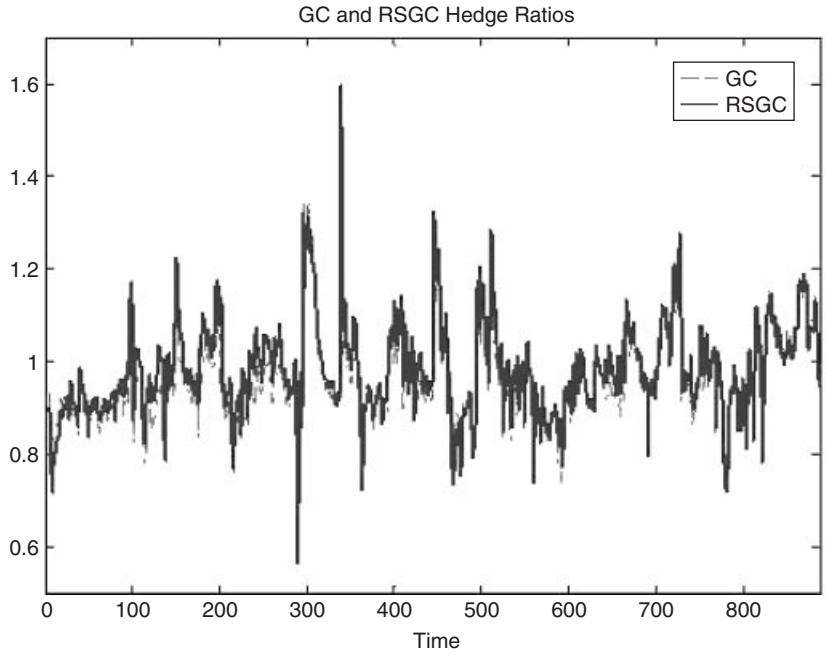


FIGURE 1
GC and RSGC hedge ratios of corn.

The term $\theta_1 + \theta_2$ in the correlation equation has the largest value for cocoa data, which is equal to 0.988 and has the smallest value for corn data, which is equal to 0.43. This shows that cocoa has the largest correlation persistence and corn has the smallest correlation persistence. Most of these parameters are significant implying the need to model the time-varying correlation of spot and futures returns. For corn data, the state-dependent shifting parameter π_1 and π_2 are equal to 0.235 and 0.385 in low- and high-volatility states, respectively. The lower tail dependence increases in the high-volatility state implies that spot and futures markets are more likely to collapse together in the high-volatility state. Wheat data have similar results. For oats and cocoa data, however, higher tail dependence increases in the high-volatility state, which implies that spot and futures markets are more likely to boom together in the high-volatility state.

Table III presents summary statistics regarding the effects of static and dynamic hedging strategies on out-of-sample hedging effectiveness. In-sample estimation was conducted using data from January 1991 to December 2006 left the final year for out-of-sample analysis. This study investigates the out-of-sample hedging effectiveness of futures hedging. For the hedger, what matters

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TABLE III
Out-of-Sample Hedging Effectiveness

	Variance of Hedged Portfolio Return	*Improvement of RSGC Over Other Hedging Strategies*[a]	*Expected Weekly Utility*[b]	*Utility Gain of Dynamic Hedging Models Over OLS*[c]
Corn				
Unhedged	27.653			
OLS	2.886	2.22%	−11.51	
CC	2.452	0.65%	−9.76	1.75
VC	2.440	0.60%	−9.70	1.81
GC	2.312	0.14%	−9.26	2.25
RSGC	2.272		−9.09	2.42
Oats				
Unhedged	19.766			
OLS	11.229	3.83%	−44.92	
CC	10.725	1.28%	−42.92	2.00
VC	11.590	5.65%	−46.24	−1.32
GC	11.101	3.18%	−44.32	0.60
RSGC	10.472		−41.88	3.04
Wheat				
Unhedged	27.953			
OLS	5.524	1.04%	−22.05	
CC	5.331	0.35%	−21.33	0.72
VC	5.488	0.91%	−22.05	0.00
GC	5.295	0.22%	−21.27	0.78
RSGC	5.234		−21.03	1.02
Cocoa				
Unhedged	8.776			
OLS	4.257	−0.60%	−16.98	
CC	4.322	0.14%	−17.29	−0.31
VC	4.355	0.51%	−17.52	−0.54
GC	4.436	1.44%	−17.84	−0.86
RSGC	4.310		−17.33	−0.35

Note. Hedging period is from January 2007 to December 2007.

[a]Improvement of RSGC over other hedging strategies is defined as the difference of the percentage variance reduction of RSGC and the percentage variance reduction of alternative hedging strategies.

[b]Expected weekly utility is calculated based on Equation (27).

[c]Utility gains of dynamic hedging models over OLS are defined as the differences of the expected utilities of alternative dynamic models and the expected utility of OLS.

most is the hedging performance in the future not in the past. The performance of RSGC is compared with performances of OLS, CC, the constant correlation GARCH (Bollerslev, 1990), VC, the varying correlation GARCH (Tse & Tsui, 2002), and GC, the state-independent Gumbel–Clayton GARCH.

Results show that RSGC hedging method exhibits good hedging performance for all data in terms of variance reduction. For corn data, RSGC has the lowest variance, which is equal to 2.727. The variance of the hedged portfolio with RSGC hedging is reduced by 2.22, 0.65, 0.6, and 0.14% compared to

OLS, CC, VC, and GC hedging. Oats and wheat data provide similar results. RSGC has the lowest variance, which is equal to 10.472 and 5.234 for oats and wheat, respectively. The variance of the hedged portfolio with RSGC hedging is reduced respectively by 3.83, 1.28, 5.65, and 3.18% compared to OLS, CC, VC, and GC hedging for oats and reduced respectively by 1.04, 0.35, 0.91, and 0.22% compared to OLS, CC, VC, and GC hedging for wheat. The RSGC for cocoa has a variance of 4.31 and is only second to OLS, which has the lowest variance of 4.257.

In general, allowing the dependence structure to be asymmetric does not always improve the hedging performance. GC has better performance than CC and VC for corn and wheat but worse performance than CC and VC for cocoa. Allowing for both asymmetric dependence and regime shift in the RSGC, however, improves the hedging effectiveness for all data. The improvement might be the fact that only RSGC captures all properties of varying correlation, state-dependent, and asymmetric dependence simultaneously compared to CC, VC, and GC.

To show the robustness of the superiority of RSGC hedging, in addition to the most recent year hedging period, the most recent two years hedging period is also examined and the results are reported in Table IV. Again, RSGC provides the highest variance reduction compared to other hedging strategies for all data investigated in this study. The data are further split into two sub-samples for investigating the consistency of the superiority of RSGC over different hedging periods. In the first sub-sample (pre-2000), in- and out-of-sample periods are from January 1991 to December 1998 and from January 1999 to December 1999, respectively, and in the second sub-sample (post-2000), in- and out-of-sample periods are from January 2000 to December 2006 and from January 2007 to December 2007, respectively. Table V presents these sub-period hedging performances. It is found that in terms of variance reduction, for the post-2000 period, RSGC has the best performance for corn and cocoa and second best performance for oats and wheat. For the pre-2000 period, RSGC has the best performance for cocoa and second best performance for oats and corn. RSGC, however, has poor performance for wheat data, which is only superior to GC. Overall, RSGC still exhibits good hedging effectiveness for longer hedging period and sub-period hedging.

Because dynamic hedging strategies are potentially more costly implement than static models due to the need for frequent rebalancing, to compare the economics value of these dynamic hedging methods, the utility improvements of these methods relative to static OLS hedging are also examined. Following other empirical studies in the literature (Alizadeh & Nomikos, 2004; Kroner & Sultan, 1993; Lafuente & Novales, 2003; Lee et al., 2006; Park & Switzer, 1995), the hedger is assumed to have an expected utility function given by Equation (27) with the coefficient of absolute risk aversion ξ equal to 4.

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TABLE IV
Out-of-Sample Hedging Effectiveness

	Variance of Hedged Portfolio Return	*Improvement of RSGC over Other Hedging Strategies*[a]	*Expected Weekly Utility*[b]	*Utility Gain of Dynamic Hedging Models over OLS*[c]
Corn				
Unhedged	23.628			
OLS	3.094	1.57%	−12.27	
CC	2.774	0.22%	−10.97	1.29
VC	2.864	0.60%	−11.33	0.94
GC	2.759	0.16%	−11.02	1.24
RSGC	2.722		−10.84	1.42
Oats				
Unhedged	15.409			
OLS	7.645	2.09%	−30.52	
CC	7.439	0.75%	−29.71	0.80
VC	8.054	4.74%	−32.20	−1.68
GC	7.873	3.57%	−31.50	−0.98
RSGC	7.323		−29.30	1.22
Wheat				
Unhedged	24.269			
OLS	6.122	0.83%	−24.46	
CC	6.368	1.85%	−25.51	−1.05
VC	6.216	1.22%	−24.91	−0.46
GC	6.056	0.56%	−24.33	0.13
RSGC	5.919		−23.74	0.72
Cocoa				
Unhedged	8.854			
OLS	3.995	2.87%	−15.93	
CC	4.141	4.52%	−16.57	−0.64
VC	3.885	1.62%	−15.63	0.29
GC	3.974	2.64%	−15.99	−0.06
RSGC	3.741		−15.06	0.87

Note. Hedging period is from January 2006 to December 2007.

[a]Improvement of RSGC over other hedging strategies is defined as the difference of the percentage variance reduction of RSGC and the percentage variance reduction of alternative hedging strategies.

[b]Expected weekly utility is calculated based on Equation (27).

[c]Utility gains of dynamic hedging models over OLS are defined as the differences of the expected utilities of alternative dynamic models and the expected utility of OLS.

As shown in Table III, for corn data, the average weekly variance of the returns from hedged portfolio for OLS and RSGC hedging are 2.886 and 2.272, respectively. Although not reported here, the hedged portfolio returns of OLS and RSGC hedging are 3.52 and 0.04%, respectively. Based on Equation (27), if an investor adopts OLS hedging, the average weekly utility is $U_{OLS} = 3.52\% - 4(2.886) = -11.51$. With RSGC, the average weekly utility is $U_{RSGC} = 0.04\% - 4(2.272) = -9.09$. The hedger's net benefit from using RSGC hedging over OLS hedging is 2.42 (242%)$-C$, where C stands for the transaction

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TABLE V
Out-of-Sample Hedging Effectiveness

	January 1999–December 1999				*January 2007–December 2007*			
	Variance of Hedged Portfolio Return	*Improvement of RSGC over Other Hedging Strategies*[a]	*Expected Weekly Utility*[b]	*Utility Gain of Dynamic Hedging Models over OLS*[c]	*Variance of Hedged Portfolio Return*	*Improvement of RSGC over Other Hedging Strategies*	*Expected Weekly Utility*	*Utility Gain of Dynamic Hedging Models over OLS*
Corn								
Unhedged	14.813				27.653			
OLS	3.407	1.35%	−13.52		2.657	1.55%	−10.59	
CC	3.275	0.45%	−12.98	0.54	2.404	0.63%	−9.56	1.03
VC	2.906	−2.04%	−11.50	2.02	2.404	0.63%	−9.56	1.03
GC	3.286	0.53%	−13.13	0.39	2.270	0.15%	−9.09	1.50
RSGC	3.208		−12.79	0.73	2.228		−8.91	1.68
Oats								
Unhedged	21.088				19.766			
OLS	12.863	5.07%	−51.39		10.997	1.39%	−43.99	
CC	12.653	4.07%	−50.57	0.82	10.471	−1.27%	−41.90	2.09
VC	11.673	−0.57%	−46.68	4.71	12.158	7.26%	−48.51	−4.52
GC	11.819	0.12%	−47.28	4.11	11.039	1.60%	−44.08	−0.09
RSGC	11.794		−47.18	4.21	10.722		−42.88	1.11
Wheat								
Unhedged	15.730				27.953			
OLS	7.243	−1.86%	−28.91		5.682	1.17%	−22.73	
CC	7.327	−1.32%	−29.27	−0.36	5.491	0.49%	−21.98	0.75
VC	7.260	−1.75%	−29.02	−0.11	5.631	0.99%	−22.40	0.33
GC	7.698	1.03%	−30.80	−1.89	5.295	−0.21%	−21.10	1.63
RSGC	7.535		−30.15	−1.24	5.355		−21.41	1.32
Cocoa								
Unhedged	20.616				8.776			
OLS	15.362	25.47%	−61.45		4.286	3.42%	−17.15	
CC	14.069	19.20%	−56.29	5.16	4.239	2.88%	−16.97	0.18
VC	11.201	5.29%	−44.68	16.77	4.103	1.33%	−16.29	0.86
GC	11.942	8.88%	−47.69	13.76	4.098	1.28%	−16.31	0.83
RSGC	10.111		−40.43	21.02	3.986		−15.93	1.22

Note. First sub-sample hedging period is from January 1999 to December 1999 and second sub-sample hedging period is from January 2007 to December 2007.

[a]Improvement of RSGC over other hedging strategies is defined as the difference of the percentage variance reduction of RSGC and the percentage variance reduction of alternative hedging strategies.

[b]Expected weekly utility is calculated based on Equation (27).

[c]Utility gains of dynamic hedging models over OLS are defined as the differences of the expected utilities of alternative dynamic models and the expected utility of OLS.

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cost from dynamic rebalancing. This implies that if $C < 242\%$, the RSGC hedging is preferred to OLS hedging. As the typical round trip transaction cost is around 0.02–0.04%, a mean-variance expected utility-maximizing hedger will benefit from hedging with RSGC even after taking account of these transaction costs. In Table III, for corn, oats, and wheat data, all dynamic hedging models generate utility gains compared to OLS method except for the VC hedging in the oats data. As for the cocoa data, all dynamic hedging models provide utility losses compared to OLS.[7] Results of utility gain or loss for two-year hedging and sub-period hedging are reported in Tables IV and V, respectively. For two-year hedging and sub-period hedging, except that OLS performs best in the wheat data in the pre-2000 period, RSGC is the only dynamic hedging model that consistently provides utility gain.

To test the statistical significance of the performance of these dynamic hedging models, DM tests are performed. Table VI presents the DM statistics, which test the superiority of alternative dynamic hedging models over conventional OLS hedging. For two-sided tests, the null of equal variance at the 5% level is rejected if $|DM| > 1.96$, where DM stands for the Diebold–Mariano statistics. As shown in Table VI, most of the DM statistics are significant. Only the full sample, one-year hedging, and pre-2000 sub-sample of VC for wheat and the post-2000 sub-sample of GC for oats cannot reject the equal variance hypothesis. For one-sided tests, the null that benchmark (OLS) is not outperform by alternative models is not rejected at the 5% level if $DM < -1.645$. For corn data, all dynamic hedging models have positive DM statistics and larger than 1.96, implying that we can reject the null that OLS is not outperformed by dynamic hedging models. No consistent results for oats, wheat, and cocoa, but we can reject the null that OLS is not outperformed by dynamic hedging models for most of the cases, especially for the RSGC model.

What we are interested is not only whether the dynamic hedging models outperform the OLS model, but also whether the RSGC outperforms the state-independent GC, symmetric dependence VC, and constant correlation CC. RSGC is the only GARCH model that possesses all properties of regime switching, varying correlation, and asymmetric dependence. The DM statistics that test the superiority of RSGC over CC, VC, and GC are reported in Table VII. For cocoa data, RSGC has positive DM statistics and larger than 1.96 except for the full sample, one-year hedging of CC, which has a DM statistics equal to 0.31. This implies that RSGC outperforms CC, VC, and GC significantly and is no worse than CC for the case of full sample, one-year hedging. For corn data, RSGC significantly outperforms CC, VC, and GC except for the case of

[7]Byström (2003) and Lee et al. (2006) also report similar results that OLS outperforms more elaborate dynamic hedging models for electricity futures traded in the Nordic Power Exchange and lead futures traded in the London Metal Exchange, respectively.

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TABLE VI

Dibold–Mariano Test Statistics of No Superiority of Alternative Dynamic Hedging Models Over Conventional OLS Hedging

	Corn				*Oats*				*Wheat*				*Cocoa*			
	Full Sample		*Sub-Sample*		*Full Sample*		*Sub-Sample*		*Full Sample*		*Sub-Sample*		*Full Sample*		*Sub-Sample*	
Hedging Period	*Jan 2007–Dec 2007*	*Jan 2006–Dec 2007*	*Jan 1999–Dec 1999*	*Jan 2007–Dec 2007*	*Jan 2007–Dec 2007*	*Jan 2006–Dec 2007*	*Jan 1999–Dec 1999*	*Jan 2007–Dec 2007*	*Jan 2007–Dec 2007*	*Jan 2006–Dec 2007*	*Jan 1999–Dec 1999*	*Jan 2007–Dec 2007*	*Jan 2007–Dec 2007*	*Jan 2006–Dec 2007*	*Jan 1999–Dec 1999*	*Jan 2007–Dec 2007*
CC	47.11	99.51	13.95	18.22	48.66	73.85	61.32	43.65	23.80	−68.71	−14.54	33.48	−29.67	127.59	112.69	12.90
VC	42.69	61.95	37.02	18.22	−9.45	−38.37	107.11	−20.58	1.76[#]	−21.38	−1.75[#]	6.66	−12.70	38.84	89.13	44.84
GC	44.07	81.71	7.47	31.91	4.59	−27.03	64.94	−1.92[#]	27.60	48.66	−59.42	48.59	−19.43	4.69	65.01	48.18
RSGC	49.82	92.26	15.04	34.82	85.07	87.61	79.98	23.99	19.31	72.76	−52.18	30.32	−12.87	57.91	75.12	44.94

Note. (1) For two-sided test, the null of equal variance at the 5% level is rejected if $|DM|>1.96$. For one-sided test, the null that benchmark is not outperformed by alternative model is not rejected at the 5% level if $DM < -1.645$. (2) [#]Indicates insignificant at the 5% level for two-sided test.

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TABLE VII

Dibold–Mariano Test Statistics of No Superiority of RSGC Over Other Competing Dynamic Hedging Models

	Corn				*Oats*				*Wheat*				*Cocoa*			
	Full Sample		*Sub-Sample*		*Full Sample*		*Sub-Sample*		*Full Sample*		*Sub-Sample*		*Full Sample*		*Sub-Sample*	
Hedging Period	*Jan 2007–Dec 2007*	*Jan 2006–Dec 2007*	*Jan 1999–Dec 1999*	*Jan 2007–Dec 2007*	*Jan 2007–Dec 2007*	*Jan 2006–Dec 2007*	*Jan 1999–Dec 1999*	*Jan 2007–Dec 2007*	*Jan 2007–Dec 2007*	*Jan 2006–Dec 2007*	*Jan 1999–Dec 1999*	*Jan 2007–Dec 2007*	*Jan 2007–Dec 2007*	*Jan 2006–Dec 2007*	*Jan 1999–Dec 1999*	*Jan 2007–Dec 2007*
CC	32.05	23.65	10.15	34.14	16.87	24.54	59.37	−34.65	1.26#	147.17	−21.54	2.96	0.31#	119.06	66.87	37.98
VC	32.83	57.17	−36.62	34.18	31.23	82.25	−25.35	22.02	28.96	77.08	−23.21	30.19	6.35	5.41	42.10	13.23
GC	9.38	15.97	12.41	8.96	30.44	87.63	4.99	9.83	−1.98	20.96	27.84	−35.22	19.11	34.09	70.18	11.52

Note. (1) For two-sided test, the null of equal variance at the 5% level is rejected if $|DM| > 1.96$. For one-sided test, the null that benchmark is not outperformed by alternative model is not rejected at the 5% level if $DM < -1.645$. (2) #Indicates insignificant at the 5% level for two-sided test.

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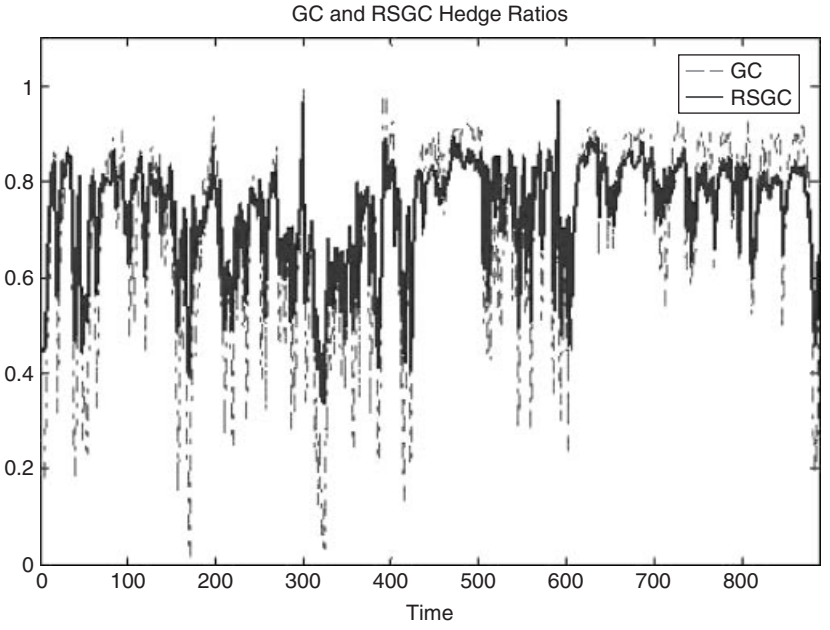


FIGURE 2
GC and RSGC hedge ratios of oats.

pre-2000 period of VC and for the oats data; RSGC significantly outperforms CC, VC, and GC except for the case of post-2000 period of CC and pre-2000 period of VC. No consistent results for wheat, but in general, RSGC performs better for more recent data for wheat.

Figures 1–4 show the hedge ratios estimated by using RSGC and GC for corn, oats, wheat, and cocoa, respectively. The conditional hedge ratios are clearly changing as new information arrives at the market and the hedge ratios are very volatile revealing that adjustment of the hedge portfolio using dynamic hedging strategies is highly required. Corn, oats, and wheat hedge ratios exhibit the property of mean reverting but this property is less clear in the cocoa data. This implies that the impact of a shock to the corn, oats, and wheat hedge ratios eventually becomes negligible and the impact of a shock to the cocoa hedge ratio is more persistent.[8]

CONCLUSIONS

The article develops a regime-switching Gumbel–Clayton (RSGC) copula GARCH model for optimal futures hedging. RSGC solves problems of recently

[8]To save space, summary statistics and the results of the unit root test on hedge ratios are not reported but available upon request. The augmented Dickey-Fuller tests on hedge ratios also support this observation.

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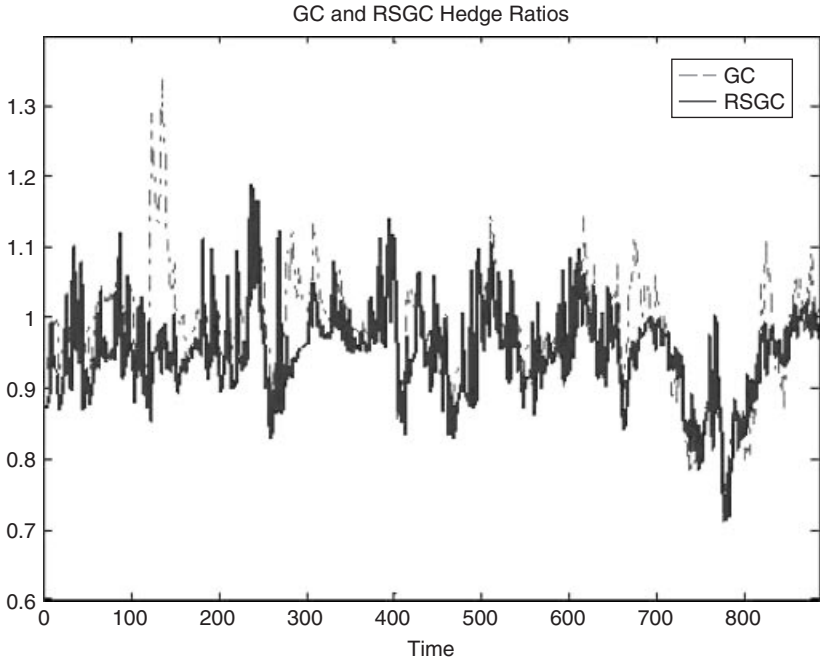


FIGURE 3
GC and RSGC hedge ratios of wheat.

developed regime-switching models. It specifies a Gumbel–Clayton copula to capture the asymmetric dependence structure and adopts an independent switching GARCH process to avoid using Gray's recombining method, which is problematic in interpreting and analyzing the variance process. In addition, based on the assumption of the independent switching, a modified formula is derived for calculating the minimum variance hedge ratio. RSGC captures properties of Markov regime switching, time-varying correlation, and asymmetric dependence structure simultaneously. This article investigates if incorporating these properties improves the futures hedging effectiveness.

Four agricultural commodities, corn, oats, wheat, and cocoa are investigated in this study. Based on nested and non-nested likelihood ratio tests, RSGC is found to perform better than the constant correlation CC, symmetric dependence VC, and state-independent GC in explaining the data. Empirical results of out-of-sample hedging effectiveness show that RSGC exhibits good hedging performance in terms of variance reduction. When full sample are used, RSGC is found to have lowest hedging portfolio variance or highest portfolio variance reduction for both one-year and two-year hedging periods. The only exception is the one-year hedging for cocoa data. In that case, the hedging effectiveness of RSGC is only second to the conventional OLS. When sub-period

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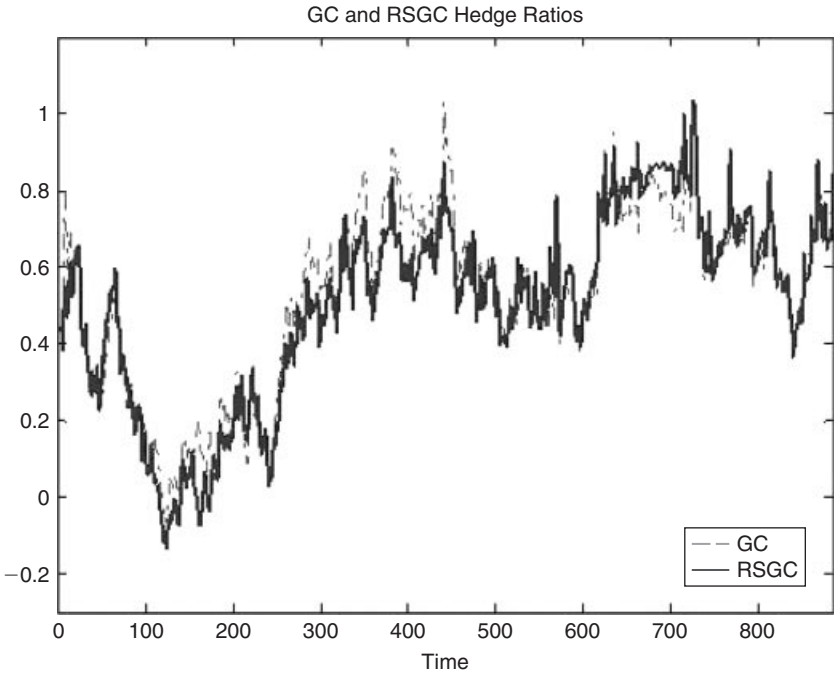


FIGURE 4
GC and RSGC hedge ratios of cocoa.

hedging is investigated, RSGC still exhibits good hedging effectiveness. RSGC performs either best or second best for all data except for wheat data in the pre-2000 period.

To better understand the economic significance of portfolio variance reduction, the utility-based criterion is used to investigate how a hedger's utility is improved with dynamic hedging methods relative to OLS. Except that OLS outperforms all dynamic hedging strategies for the pre-2000 sub-sample of wheat and full sample one-year hedging of cocoa, only the proposed RSGC, among all dynamic hedging strategies, can consistently provide utility gain compared to the static OLS hedging. To test the statistical significance of the improvements of these dynamic hedging models, DM tests are performed. For corn data, all dynamic hedging models are superior to the conventional OLS but no consistent results for oats, wheat, and cocoa data. Again, except for the pre-2000 period of wheat and full sample one-year hedging of cocoa, only RSGC consistently reject the null of no superiority of dynamic hedging over the conventional OLS hedging. The tests of superiority of RSGC over CC, VC, and GC are also performed. Results show that when full sample is used, RSGC significantly outperforms CC, VC, and GC except for the GC one-year hedging

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of wheat and for the sub-period hedging, RSGC still shows good performance for most of the cases.

In conclusion, the contribution of this study is twofold. The proposed RSGC possesses properties of regime switching, varying correlation, and asymmetric dependence. It provides a very general framework for modeling dependence structure of asset returns. Furthermore, with more flexibility in the distribution specifications, the copula-based regime-switching varying-correlation GARCH model performs more effectively in futures hedging.

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